Broadband High Efficiency Quasi-Continuous Class-J Power Amplifier

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***Abstract*— In this paper, a broadband high efficiency power amplifier (PA) with quasi-continuous Class-J mode is proposed. Specifically, the continuous current waveform can be obtained within the entire operating frequency band. For lower frequency, its load condition satisfies the requirement of Class-J PA mode. While for the center frequency and higher frequencies, the fundamental termination is more closely to real axis, which extends the bandwidth. Meanwhile, the current dip can be generated by the nonlinear output capacitors, which reshapes the current waveform thus enhancing the efficiency. The proposed PA shows a relatively high performance of greater than 40 dBm saturated output power and delivers an average of 58 % drain efficiency (DE) within the frequency band of 3.5 to 5.5 GHz.**

***Index Terms*—Broadband, high efficiency, quasi-continuous Class-J, nonlinear output capacitors, power amplifier.**

I. Introduction

With the development of 5G wireless communication, the requirements for broadband highly efficient power amplifiers (PAs) are significantly increasing. The characteristic of high output power is suitable for base station and radar detection systems.

The conventional Class-J mode in single-ended PAs had been proposed [2]. It can theoretically realize the same efficiency of 78.5% as Class-B when only the pure imaginary part of the second harmonic termination is considered. Therefore, the continuous PA modes and some extended theories are revealed in [3]-[4], which still can't solve the issue and it is difficult to ensure the resistive parts of the harmonics are removed. While the frequencies in other studies are quite low thanks to the harmonic control [5]-[6].

In this paper, a quasi-continuous Class-J PA with output matching network (OMN) of microstrip multi-stub is proposed. Due to the fundamental load terminations are located nearby the real axis, which is easier to extend the bandwidth by reducing the Q-factor of OMN. Specifically, the imaginary parts of the fundamental load terminations at center frequency and higher frequencies are smaller than the terminations at lower frequencies. Therefore, the load terminations at center frequency and higher frequencies are different from the conventional Class-J PA mode. In addition, the second harmonic termination used as nonlinear output capacitors has been explored in terms of the time-domain current and voltage waveforms. For the clipping contours of

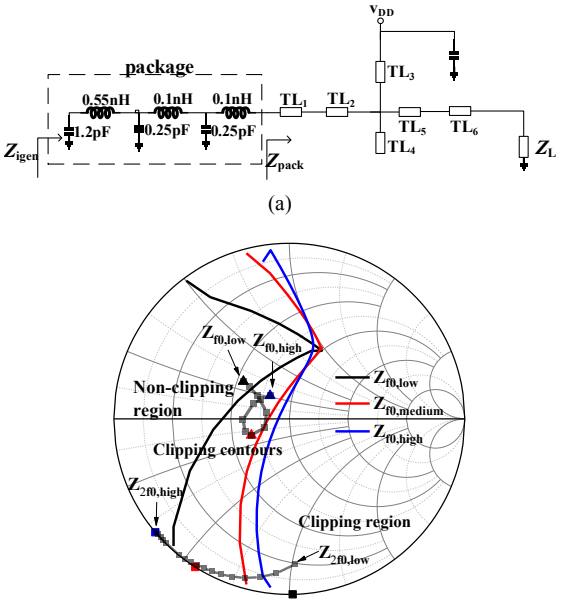


Fig. 1 Proposed OMN design. (a) The OMN circuit diagram. (b) The simulated fundamental and harmonic terminations (gray trace) at intrinsic current generator (igen) plane. Clipping contours (colored trace) dividing the clipping region and non-clipping region describing "zero-crossing" for the fundamental and second harmonic terminations.

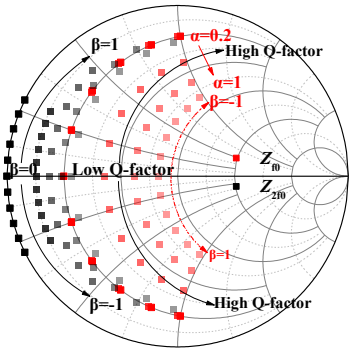


Fig. 2. Design space for the fundamental and second harmonic terminations of the resistive-reactive Class-J PA mode for α = [0.2,1] and β= [-1,1] in steps of 0.2 when γ= 0 and 0.2, respectively.

the fundamental termination, its shape is mainly determined by the specific second harmonic termination thus affecting the current waveform. Therefore, after eliminating this effect, the role of nonlinear capacitors is particularly obvious by comparing high-power and low-power regions. With these

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strategies, the broadband and highly efficient power amplifier can be realized.

II. Analysis of Realizing Broadband And High Efficiency Power Amplifier

As shown in Fig. 1(a), the OMN of the proposed quasi-continuous Class-J PA that consists of 6 microstrip lines and the packaged transistor parasitic network. In Fig. 1(b), the fundamental termination at the intrinsic current generator plane shows a small circle around real axis in the Smith chart. The second harmonic termination trajectory gets closer to the edge of the Smith chart with a small reactance as the frequency increases. As is generally known, the Q-factor, the real and imaginary parts of the fundamental loads (i.e., RL, XL), have a close relationship with the factional bandwidth (FBW), which can be expressed as [7]:

$$Z_L = R_L + jX_L \tag{1}$$

$$Q = \frac{|X_L|}{R_L} = \frac{\omega_0}{\omega_+ - \omega_-} = \frac{\omega_0}{FWB} \tag{2}$$

where ω_+ and ω_- delimit the frequency band.

Through intuitive reasoning from the above procedure, the Q-factor of the proposed fundamental termination trajectory for $-0.3<\beta<0.3$ is lower than the Q-factor of conventional Class-J/J* PA mode termination for $-1<\beta<1$ in Fig. 2. Hence, the proposed quasi-continuous Class-J PA mode is more likely to have wider bandwidth.

Before drawing the fundamental clipping contours, the algorithm is calculated against in [8]. According to [9], the resistive-reactive Class-J PA mode has brought more freedom of fundamental and second harmonic terminations, as shown in Fig. 2. Therefore, the defined voltage waveform of the quasi-continuous Class-J PA mode at intrinsic current generator drain plane is calculated as:

$$v_J(\theta) = (1-\alpha\cos\theta)(1-\beta\sin\theta)(1+\gamma\cos\theta) \tag{3}$$

and refer to the normalized voltage formula,

$$\begin{aligned} v_n(\theta) = 1 &- v_{1r}\cos\theta - v_{1q}\sin\theta - v_{2r}\cos 2\theta \\ &- v_{2q}\sin 2\theta - v_{3r}\cos 3\theta - v_{3q}\sin 3\theta \end{aligned} \tag{4}$$

Combining (3) and (4), the voltage parameters v_{1r}, v_{1q}, v_{2r}, v_{2q}, v_{3r} and v_{3q} in (4) with weighting factors α, β and γ are proportional to the corresponding termination, respectively.

There are several steps to plot the clipping contours dividing line that separates the two areas (i.e., clipping region and non-clipping region) can be obtained.

Firstly, with knowledge of the assumption of the Class-B PA mode conduction angle and the assumed sinusoidal drain current waveform. V_k, V_{dc} and I_{peak} are set to 0 V, 28 V and 186 mA, respectively. Then the R_{opt} is calculated as 30 Ω.

Secondly, for the situation where the voltage just grazes zero and the most qualified angle θ varying from $-\pi$ to π, the zero-crossing behavior should be satisfied as:

$$v(\theta) = 0;\ v'(\theta) = 0 \quad . \tag{5}$$

From (4) and (5), the system can be expressed as:

$$\begin{cases} 1 - v_{1r}\cos\theta - v_{1q}\sin\theta - v_{2q}\sin 2\theta - v_{3q}\sin 3\theta = 0 \\ v_{1r}\sin\theta - v_{1q}\cos\theta - 2v_{2q}\cos 2\theta - 3v_{3q}\cos 3\theta = 0 \end{cases} \tag{6}$$

where

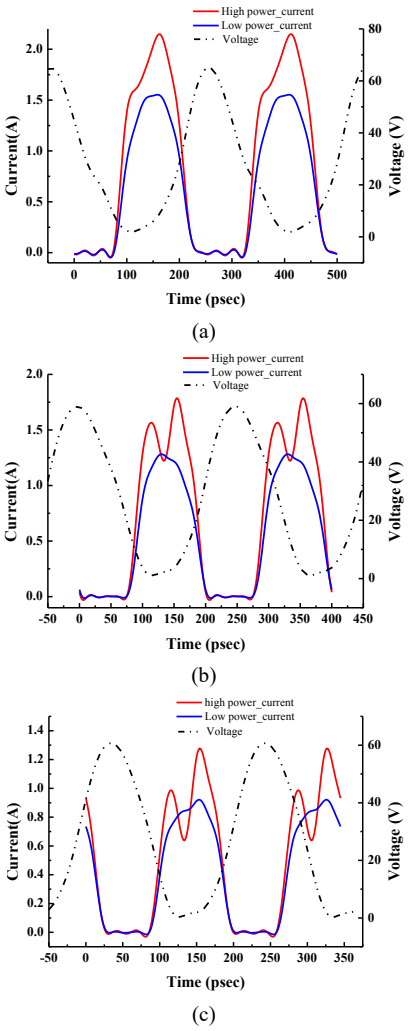


Fig. 3. Simulated current and voltage waveforms of the intrinsic current generator plane when in high power region (red current line) and low power region (blue current line). (a) 3.5 GHz. (b) 4.5 GHz. (c) 5.5 GHz.

$$\begin{cases} v_{1r} = (1 - v_{2q}\sin 2\theta - v_{3q}\sin 3\theta)\cos\theta \\ \quad + (3v_{3q}\cos 3\theta + 2v_{2q}\cos 2\theta)\sin\theta \\ v_{1q} = (1 - v_{2q}\sin 2\theta - v_{3q}\sin 3\theta)\sin\theta \\ \quad - (3v_{3q}\cos 3\theta + 2v_{2q}\cos 2\theta)\cos\theta \end{cases} \tag{7}$$

By fixing the second harmonic terminations (i.e., v_{2q}) with varying the fundamental termination around the Smith chart, the clipping contours are finally drawn as shown in Fig. 1(b). The clipping contours divides into two regions, with non-clipping region (the left region) and clipping region (the right region). The second harmonic termination trajectory (blue points) goes in reverse with optimal fundamental termination trajectory (dotted line) along the edge of the Smith chart. Based on these contours, the fundamental terminations just fall in the non-clipping region, which eliminates the impact of the second harmonics on the fundamental terminations in the clipping region.

In fact, the efficiency of Class-J PA mode can be enhanced due to the influence of the resistive parts of the fundamental terminations [10]. According to Fig. 1(b), therefore, the fundamental load terminations at center frequency and higher frequency (i.e., 4.5 and 5.5 GHz) located nearby the real axis affect the output power and drain efficiency. As

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shown in the following figures, Figs. 3(a)-(c) depict the simulated current and voltage waveforms of the intrinsic drain generator plane at 3.5, 4.5 and 5.5 GHz.

For less than 32 dBm of the low power region, the nonlinear C_{out} can be equivalent to the linear capacitor, and the characteristic of current and voltage waveforms is the shape of half-sinusoidal (blue line), which is similar as Class-B PA mode. But for the high power region, above 32 dBm, the circuit has strong nonlinear (C_{out}) effect and the nonlinear C_{out} gradually increases as the output power increases. Owing to the second harmonic components in Fig. 1(b), the current and voltage waveforms are more like the waveforms of Class-F^{-1} PA mode. In addition, they have a large dip at the middle (red line) and the current peak value increases as well, as shown in Figs. 3(b)-(c). At this moment, the nonlinear C_{out} can provide larger output power and higher efficiency.

III. Experiment Results

Based on the design method discussed in section II, a quasi-continuous Class-J PA prototype is fabricated and measured using GaN HEMT device CG2H40010F provided by CREE. The substrate is realized on a thickness of 20 mil and a permittivity of 2.2 with Rogers 5880. The drain bias is set to 28 V, and the gate bias is -2.9 V. Fig. 4 shows the photograph of the fabricated PA and the measured and simulated efficiency within the entire frequency range of 3.5 and 5.5 GHz. Table I compares this work with other PAs. The proposed PA achieves the highest operating frequency and FE, comparable power and efficiency performance compared with the other PA works.

TABLE I
COMPARISON WITH STATE-OF-THE-ART PAS

work	bandwidth (GHz) (%)	Pout (W)	DE(AE) (%)	FE*
[3]	1.3-2.4 (59)	10.2-13.1	63-72 (67)	78.1
[4]	1.6-2.4 (40)	7.4-9.3	--	--
[5]	1.5-2.5 (50)	8.9-11.2	60-70 (65)	77.3
[6]	1.65-2.7 (48)	11.2	55-72 (63.5)	77.1
This work	3.5-5.5 (40)	8.7-12.6	53-65 (59)	85.9

(FE*=AE*$\sqrt[4]{f_0}$, AE: Average DE, f_0: Center frequency, --: the data was not provided)

IV. Conclusion

In this work, a quasi-continuous Class-J power amplifier with high efficiency is proposed. This approach can extend the bandwidth by optimizing the fundamental termination by reducing the Q-factor. After getting rid of the impact of the second harmonics on the fundamental terminations in the clipping region, the Class-F^{-1}-like waveform in high power region with the second harmonics is represented. And the measurement results showed good output power and drain efficiency, which validate an effective way to achieve good performance.

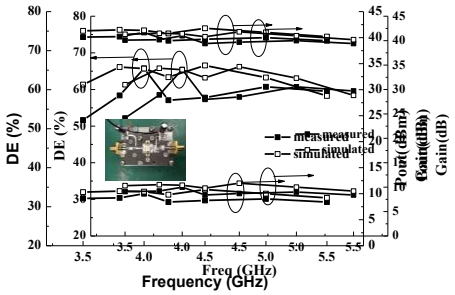


Fig. 4. The fabricated PA, measured and simulated results from 3.5 to 5.5 GHz.

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